

# Weight perturbation across four AI weather models

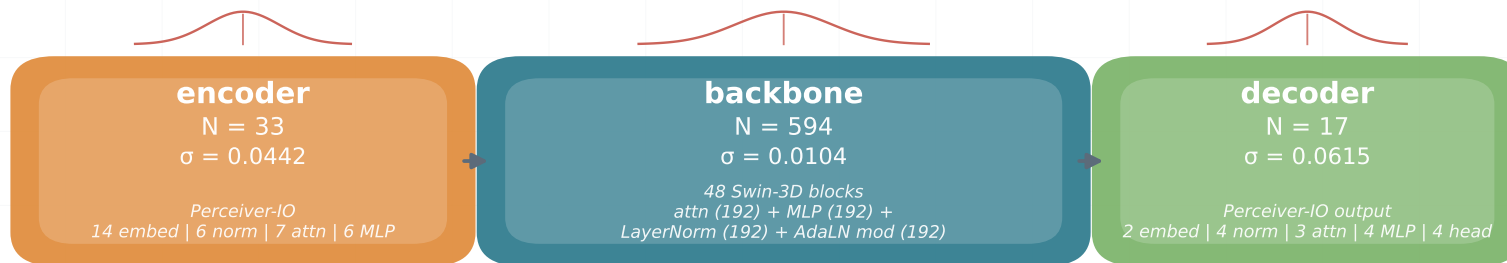
Phase 2 layer-group ablation at  $\sigma_{\text{full}} = 0.01$  with  $\sqrt{N_{\text{total}}/N_{\text{group}}}$  variance scaling. Multiplicative noise:  $w' = w (1 + \sigma \cdot N(0,1))$ .



## Aurora

Swin-Transformer 3D U-Net

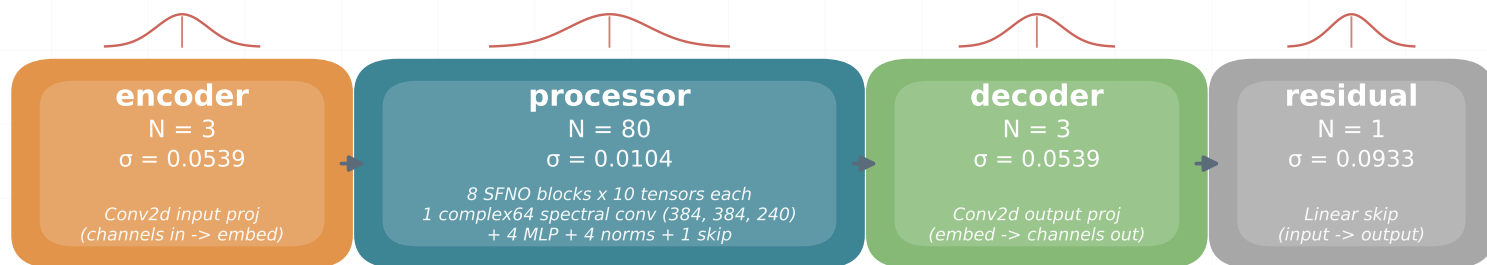
$N_{\text{total}} = 644$



## SFNO

Spherical Fourier Neural Operator

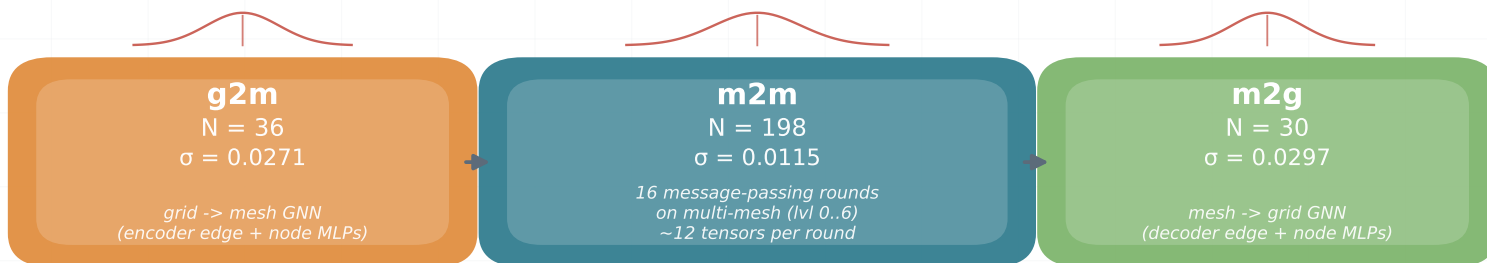
$N_{\text{total}} = 87$



## GraphCast

Multi-Mesh Graph Neural Network

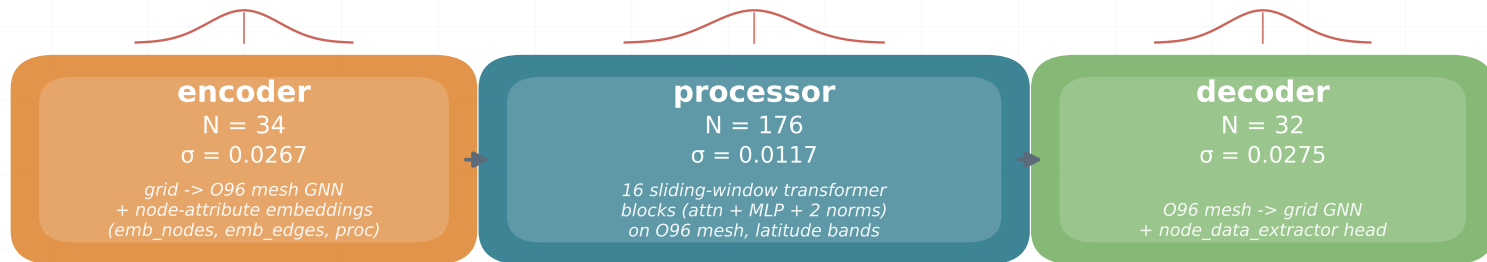
$N_{\text{total}} = 264$



## AIFS

GNN + latitude-band transformer

$N_{\text{total}} = 242$



**encoder** **processor** **decoder** **aux / residual**  = Gaussian noise injection on every learnable tensor in the group